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CLASSIFICATION PROJECT

Home Advantage in Football Football

Bias, or real?

Team
Group 5

Method
Decision Tree Classifier

Sample
50,000 matches

THE QUESTION

Home teams win 47% of matches. Away teams win 29%.

Is that gap the crowd, the travel and the referee — or simply stronger teams happening to play at home?

THE DATA

The dataset in numbers

50,000

matches

one row per match

25

columns

22 numeric, 3 text

8

leagues

EPL most, Eredivisie fewest

13

pre-match features

the model's only inputs

0

missing values

no imputation needed

0

duplicate rows

match_id is unique throughout

THE SAMPLE

Eight leagues, one pattern

EPL
England · most

La Liga
Spain

Serie A
Italy

Bundesliga
Germany

Ligue 1
France

Eredivisie
Netherlands · fewest

Primeira
Portugal

Championship
England, tier 2

46–47% home win rate in every one of the eight. The spread is about a single percentage point, so the competition a match is played in barely predicts its result.

FEATURE AUDIT

Twenty-five columns, four groups

Group	Example columns	n	Model input
Team strength	elo_home, elo_away, elo_diff	3	Yes
Crowd conditions	crowd_present, attendance_pct …	4	Yes
Travel & context	travel_distance_km, is_derby …	6	Yes
In-match stats	fouls_home, yellow_away, red_home …	6	No — in-play only
Outcome	goals_home, goals_away, home_points …	4	No — leakage
ID & target	match_id, home_result	2	Dropped / predicted

13
columns used as model inputs

11
columns dropped before training

1
target: home_result

DATA LEAKAGE

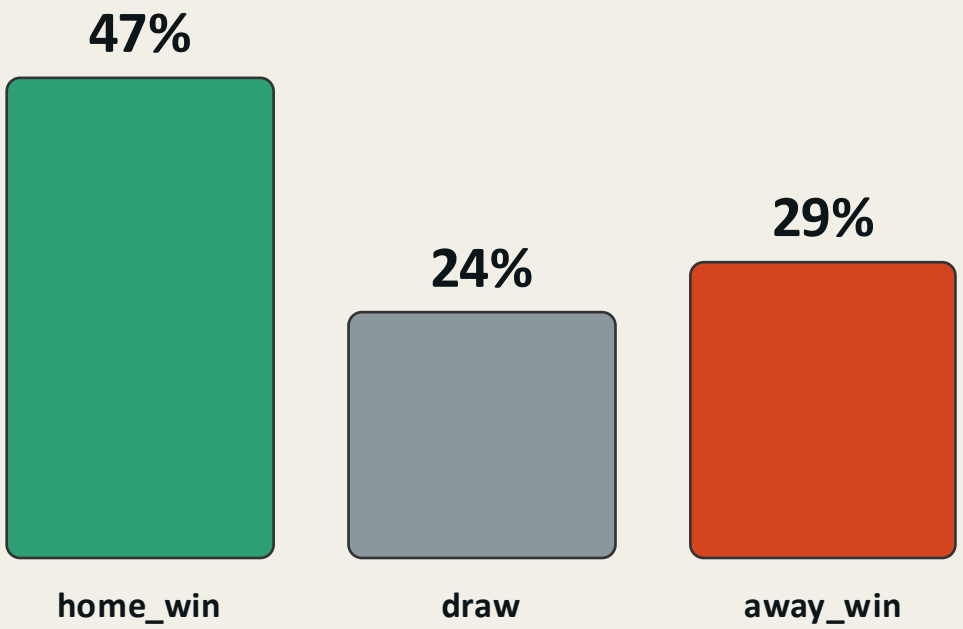
Eleven columns left out on purpose



Goals, fouls and cards are only known once the match is under way. A model given those columns would score beautifully and predict nothing useful — so every in-match and outcome column was removed before training.

THE TARGET

What the model predicts



Three outcomes, seen from the home team’s side, and the classes are imbalanced. Draws are the smallest at 24% and track no single feature cleanly, which makes them the hardest class for any classifier to recover.

BASELINE TO BEAT

46.9%

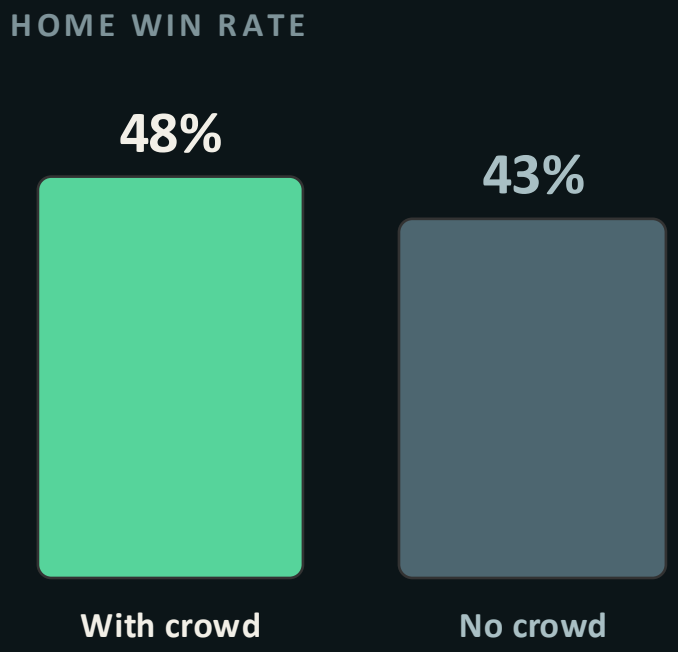
accuracy from always guessing home_win. Accuracy alone can mislead here, so macro F1 and a per-class report are reported too.

FEATURE · CROWD_PRESENT

Matches without a crowd

≈20%

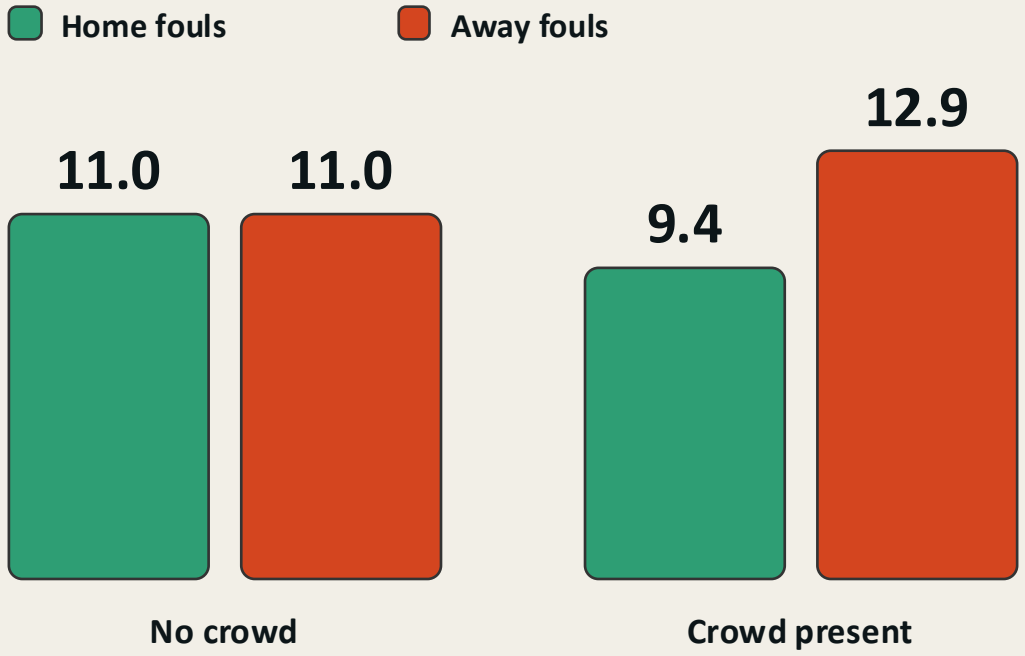
of matches were played in an empty stadium. Both attendance_pct and crowd_size spike hard at zero — the sample deliberately carries “ghost games”, which is what makes the crowd’s effect measurable at all.



A crowd is worth about 5 percentage points — real, but modest.

THE MECHANISM

Fouls and cards under crowd pressure



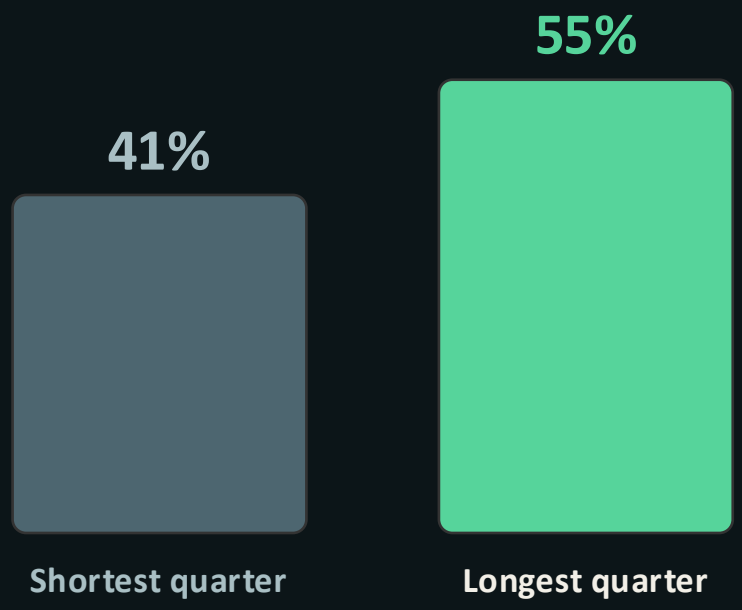
With an empty stadium the two sides are punished almost identically — about 11 fouls each. Put a crowd in and the gap opens: the away side concedes more fouls and collects more yellow cards.

These columns never enter the model, because they are only known during play. They matter because they point at **why** home advantage exists: officiating shifts under crowd pressure.

FEATURE · TRAVEL_DISTANCE_KM

Travel distance

HOME WIN RATE BY THE AWAY TEAM'S TRIP



The further the away side travels, the more often the home side wins. The rate climbs across all four distance quarters — a 14-point swing, and the largest effect in the data outside team strength.

≈2,400 km

longest trip in the sample. The column is right-skewed: most journeys are short, with a long thin tail.

+0.11

correlation with a home win — fourth strongest of any column.

FEATURE · TIMEZONES_CROSSED

A feature with almost no variance



Share of all 50,000 matches by the number of time zones the away team crossed. The third band is drawn at a visible minimum — at true scale it would be a fraction of a pixel.

≈46,100

crossed 0 time zones

roughly 92% of the sample

≈3,900

crossed 1 time zone

roughly 8% of the sample

2

crossed 2 time zones

two matches in fifty thousand

The tree is allowed to split on this column, but with 92% of rows sitting in a single bucket there is almost nothing to split on — and two matches cannot support a conclusion about anything. Worth keeping in the deck precisely because it looks like a feature and behaves like a constant.

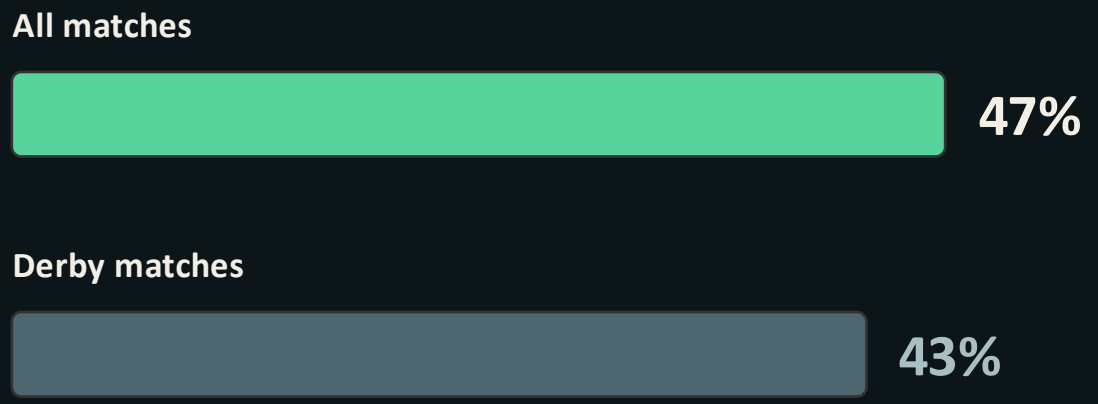
FEATURE · IS_DERBY

Derby matches

≈6%

of fixtures are local derbies — a small slice of the sample, and the one place where home advantage visibly thins out.

HOME WIN RATE

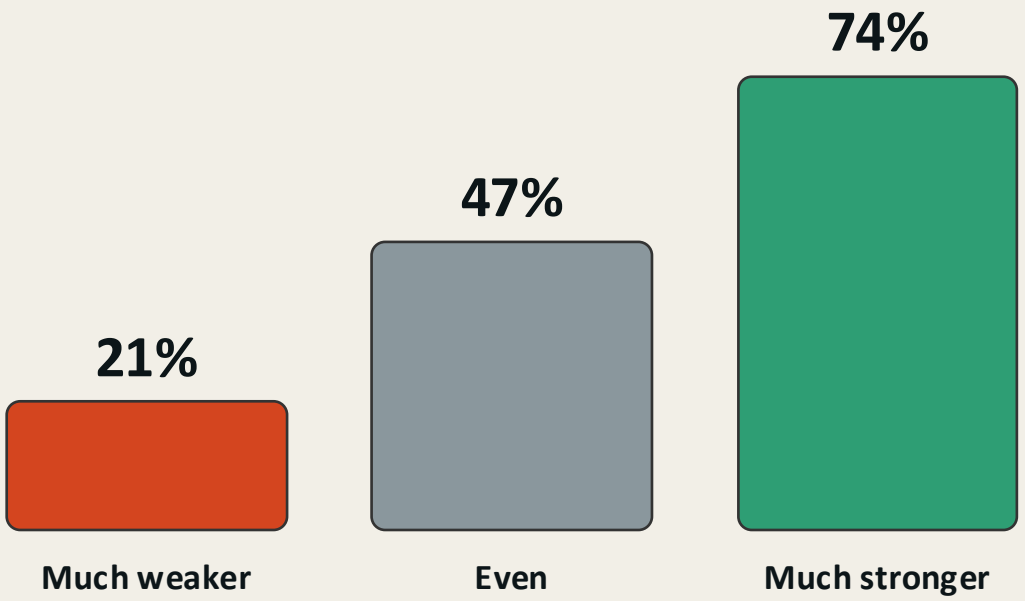


Rivals travel almost nothing and know the ground. Strip travel and unfamiliarity out of a fixture and roughly four points of the advantage go with them.

FEATURE · ELO_DIFF

Team strength

HOME WIN RATE BY STRENGTH GAP



The single strongest predictor in the dataset. A home side 300 Elo points clear wins roughly three times as often as one 300 points behind.

1200 – 1900

Elo range for both sides. elo_diff is centred on zero, so the two teams are evenly matched on average.

+0.25

correlation with a home win — more than double the next column.

THE FINDING

Equal strength, unequal result

46.5%
home team wins

28.9%
away team wins

17.6
point gap

Those are the rates among matches where the two sides are **evenly matched on Elo**. With team strength held level, the gap that remains is home advantage itself.

Matches with elo_diff between −100 and +100

RANKING THE COLUMNS

Correlation with a home win



Bars show magnitude; the sign is in the number. A stronger away side pushes the home win rate down, which is why elo_away is negative. Correlation only catches **linear** relationships — a decision tree also picks up thresholds and interactions, such as travel combined with a crowd, which is part of why it suits this data.

REDUNDANCY

Features built from other features

EXACTLY DERIVED

elo_diff = elo_home − elo_away

ALMOST EXACTLY DERIVED

crowd_size ≈ stadium_capacity × attendance_pct

Neither pair carries information its parents do not already hold. A decision tree is untroubled — it splits on whichever column separates the classes best and leaves the duplicates alone. A linear model would have had to drop one of each pair first.

METHOD

The model

INPUTS

13 pre-match features

Everything knowable before kick-off, and nothing else.

PREPROCESS

ColumnTransformer

One-hot encoding on league; the 12 numeric columns pass straight through.

CLASSIFY

Decision Tree

Predicts home_win, draw or away_win for each match.

40,000
training matches

10,000
held back for testing

`criterion = gini` `max_depth = 5` `min_samples_split = 10` `min_samples_leaf = 5` `class_weight = balanced` `random_state = 42`

80/20 split, stratified on home_result · depth capped at 5 to keep the tree readable

THEORY

How the tree picks a split

Gini impurity

$$G = 1 - \Sigma\, p^2$$

Zero when every sample in a node belongs to one class, largest when the classes are evenly mixed. The scikit-learn default, used here.

Entropy

$$H = -\Sigma\, p \log_2 p$$

A measure of disorder in a node, also zero for a pure one. The alternative criterion, selected with criterion = “entropy”.

Information gain

$$IG = H(\text{parent}) - \Sigma\, (N_j / N)\, H(j)$$

The fall in impurity a split achieves, each child weighted by its share of the samples. The tree takes whichever split maximises it.

The tree repeats this at every node until it hits a stopping rule — depth five, ten samples to split, five in any leaf. Each leaf predicts the majority class among the matches that reach it.

EVALUATION

Model versus baseline

BASELINE · ALWAYS PREDICTS HOME_WIN

Accuracy

46.9%

Macro F1

0.21

It never predicts a draw or an away win at all, so two of the three classes score zero and macro F1 stays low.

DECISION TREE · DEPTH 5, BALANCED

Accuracy

47.0%

Macro F1

0.42

Balanced class weights make the tree attempt all three classes — it trades easy home-win guesses for coverage of the other two.

Accuracy barely moves, but macro F1 doubles — and that is the whole story. The baseline never attempts a draw or an away win; the tree predicts all three. Per class: home wins F1 0.59, away wins 0.46, draws 0.21.

CONCLUSIONS

What the analysis concludes

46.5% vs 28.9%

Home advantage is real, and it survives between evenly matched sides.

+0.25

elo_diff is the strongest column by more than double. Team strength dominates.

+5 points

A crowd lifts the home win rate from 43% to 48% — real, but a minor share.

41% → 55%

The further the away side travels, the more often the home side wins.

43%

In derbies the advantage thins — short trips, familiar ground, four points lost.

24%

Draws follow no single feature and remain the hardest class to predict.

IN ONE LINE

Real, but mostly the better team

Playing at home is worth a genuine few points. Being the stronger side is worth a great deal more.



Team

Group 5

Dataset

Kaggle · sergionefedov

Code

Project repository